INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

FORWARD AREA REFUELING EQUIPMENT

(FARE)

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable fuel-dispensing system for use in forward areas to refuel aircraft, ground vehicles, and other powered equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1899	-	Cushioning Material, Packaging (Cellulosic, Water-Resistant).
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin.
P-D-680	-	Dry Cleaning Solvent.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-P-40	-	Packaging and Packing of Hand Tools.
PPP-T-60	-	Tape: Packaging, Waterproof.
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA, 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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MIL-P-116	-	Preservation, Methods of.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-G-3056	-	Gasoline, Automotive, Combat.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-N-5877	-	Nozzle, Pressure Fuel Servicing, Locking, Type D-1, and D-2, Nominal 2-1/2 Inch Diameter.
MIL-S-7916	-	Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant.
MIL-F-46162	-	Fuel, Diesel, Referee Grade.
MIL-C-46168	-	Coating Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-F-52556	-	Filter-Separator, Liquid Fuel: Frame Mounted, 100 GPM Capacity.
MIL-P-52746	-	Pump Assembly, Flammable Liquid, Centrifugal, Self-Priming, 2-Inch, 100 GPM at 100 Feet Total Dynamic Head.
MIL-N-52747	-	Nozzle Assembly, Closed-Circuit Refueling.
MIL-N-52748	-	Nozzle, Adapter, Closed-Circuit to Gravity Fill.
MIL-C-83413	-	Connectors and Assemblies, Electrical, Aircraft Grounding.
MIL-C-83413/7	-	Connectors and Assemblies, Electrical, Aircraft Grounding: Grounding Clamp Connector for Type I and III Grounding Assemblies, Clip, Electrical.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	-	Lubrication of Military Equipment.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems; Equipment, and Facilities.
MIL-STD-1474	-	Noise Limits for Military Material.
MS25384	-	Electrostatic Discharger Jumper, Fuel Nozzle-to-Aircraft.
MS70091	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Flanged, Type VIII, (For D-1 Nozzle - Aircraft).

HANDBOOK

MILITARY

MIL-HDBK-113	-	Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for use in Ground Equipment Systems.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13219E0500 - Forward Area Refueling Equipment.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSP, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1655 - Aviation Turbine Fuel.
D 3699 - Kerosine.
D 3953 - Strapping, Flat Steel and Seals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding - Steel.
D1.2 - Structural Welding - Aluminum.

(Applications for copies should be addressed to the American Welding Society, Inc. 2501 Northwest Seventh Street, Miami, FL 33125.)

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ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Department of Defense Materiel On Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 1920 L Street, NW, Washington, DC 20036.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road., Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 S. Riverside Plaza, Chicago, IL, 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The forward area refueling equipment, hereinafter referred to as FARE, shall be as shown on TA13219E0500 and as specified herein. The major components of FARE include the pump-engine assembly, two discharge hose and fitting kits, and two suction hose and ground rod kits.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the FARE. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

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3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3. Any changes or deviations of the FARE system from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish FARE systems conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The FARE system shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the FARE may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Cadmium plated fasteners. Unless deemed absolutely necessary (see 6.2 and 6.8), cadmium plated fasteners shall not be used.

3.3.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the FARE system may be newly fabricated from recovered materials to the maximum extent practicable, provided the FARE systems produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the FARE systems.

3.3.6 Rubber components. All rubber items shall not be more than 12 months old on date of acceptance by the Government.

3.4 Threads and threaded connections. All pipe and machine screw threads shall conform to FED-STD-H28. Prior to assembly, threads on all screwed fittings shall be coated with sealing compound conforming to MIL-S-7916.

3.5 Components. The FARE when assembled for field movement shall have all components as specified herein securely contained within the frame assemblies as shown in TA13219E0500.

3.6 Pump assembly. The pump assembly shall be in accordance with MIL-P-52746.

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3.7 Filter-separator. The filter-separator assembly shall be in accordance with MIL-F-52556.

3.8 Nozzle. The closed-circuit refueling nozzle shall be in accordance with MIL-N-52747.

3.8.1 Single point nozzle. The single point refueling nozzles conforming to MIL-N-5877, type D-1, shall be provided with each FARE. The nozzle shall be equipped with the following options: Non-breakaway collar and handles with bumper and dust caps, 100 mesh filtration screen, grounding cable as described in 3.8.1.1, long handle grips, and with a 2-inch female cam-lock conforming to MS70091.

3.8.1.1 Ground cable assembly. The nozzle assemblies shall be equipped with a ground cable assembly for grounding the nozzle to the vehicle and to a ground stake. The nozzle-to-vehicle ground cable shall conform to MS25384, except the clip shall conform to MIL-C-83413/7. The nozzle-to-ground stake cable shall be coated wire 15 feet long, and the end shall be equipped with a clamp in accordance with MIL-C-83413/7, suitable for bonding the cable assembly to a 5/8-inch diameter ground stake.

3.9 Nozzle adapter. The nozzle adapter shall be in accordance with MIL-N-52748.

3.10 Environmental.

3.10.1 Operating temperature. The FARE shall perform as specified herein in any ambient temperature from +125 to -25 °F.

3.10.2 Storage temperature. The FARE shall not be damaged by storage in any ambient temperature from +160 to -50 °F.

3.11 Reliability. The lower test Mean-Time-Between-Failure (MTBF) shall be 75 hours used and the upper test MTBF shall be 150 hours. The FARE shall be tested as specified in 4.6.3.

3.12 Maintenance ratio. The FARE shall have a maintenance ratio not greater than 0.08.

3.13 Performance.

3.13.1 Capacity. The FARE (when assembled as specified for operation, and tested as specified in 4.6), shall deliver not less than 50 gallons per minute (gpm) simultaneously through each of two nozzles specified in 3.8 with nozzle adapters specified in 3.9.

3.13.2 Compatibility. The FARE components shall be compatible when connected and shall not leak when tested as specified in 4.6.5.2.

3.13.3 Electrical continuity. The FARE shall be equipped to provide complete electrical continuity throughout the system. The maximum resistance from the outlet

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of the fuel source to the nozzle shall not exceed 50,000 ohms when tested using an ohmmeter. Ground wire equipped with clamps conforming to MIL-C-83413/7 shall be provided.

3.14 Lubricants. The procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL) the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.15 Treatment and painting. The portions of the FARE normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G as applicable. CARC paint shall not be applied to surfaces exceeding 400 °F.

3.16 Transportability. The FARE, prepared for shipment, shall be suitable for transport by rail. The items shall meet the test requirements of Association of American Railroads (AAR) recommended procedures for conducting field impact tests of loaded freight cars. The package FARE shall be rail transportable in CONUS and NATO countries without restrictions. The FARE shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with AAR outline diagram when loaded on a 50-inch (127 cm) high rail car. The FARE shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.6.7 without failure, damage, or permanent deformation.

3.17 Instructions. An instruction plate/placard shall be provided with a diagram showing the assembly of the FARE components and a brief description of the assembly procedure. The instruction plate/placard shall be attached in a conspicuous location.

3.18 Workmanship.

3.18.1 Castings and forgings. All parts, components, and assemblies of the FARE which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.18.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid over-heating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.18.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered.

3.18.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, other foreign matter, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the FARE to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875-inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the FARE shall be at the option of the contractor. Welding procedures shall be qualified in accordance with applicable ASME or AWS codes, and all welding and welds shall be in accordance with ASME or AWS code.

3.18.5 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operations to be performed and that such qualification is effective as defined by the particular code: AWS D1.1, and AWS D1.2, and ASME Boiler and Pressure Vessel Code, Section IX. Welders who make only horizontal welds need not qualify welds for "all position welding." The certification shall be made available for review by the contracting officer or designated representative upon request. The Government reserves the right to require the recertification of any welder or welding operator.

3.18.6 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Bolt holes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued and shall have full thread engagement. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.18.7 Riveting. Rivet holes in the sizes recommended by standard practice shall be accurately punched or drilled, and shall have all burrs removed. Rivets shall be neatly set, concentric with rivet holes, and in full contact with the surfaces of the members.

3.19 Safety. All rotating or reciprocating parts shall be guarded so as not to be a hazard to operating personnel. The engine exhaust system shall be located and installed so as to minimize the hazard of fire in the event of fuel spillage during refueling or leakage from fuel dispensing hoses and connections. The exhaust system shall incorporate a spark arrestive muffler. An emergency shut-off device shall be

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located at the pump to quickly and completely shut off the flow of fuel in case of an emergency.

3.20 Human factors engineering. The FARE system and associated equipment shall comply with the applicable design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to, MIL-STD-1472, 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 Labeling, 5.9 (Design for Maintainer), and 5.13 (Hazards and Safety) as applicable.

3.21 Noise limits. The noise produced by the FARE shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474 5.2, 5.3, and 5.4, when tested in accordance with MIL-STD-1474 and 4.6.1.8. The provisions of MIL-STD-1474, 4.2 and 4.3, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the Government contracting officer and written permission to exceed the 85 Db(A) limit is obtained from the Government contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Acceptability criteria. FARE systems which conform to all requirements in section 3 and 5 and pass all applicable examinations and tests in section 4 of this document will be considered acceptable by the Government.

4.1.3 Component and material inspection. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7).

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The contractor is responsible for ensuring that components and materials are manufacturing, examined and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.4 Disassembly inspection. Failure of any examination or test shall be cause for disassembly, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced specifications, standards, and drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with a such requirements shall be rejected and shall be cause for system rejection. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government furnished property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3 and 4.8.1).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article unit(s) shall be examined for the defects specified in 4.5. Presence of one or more defects shall be cause for rejection of the first article.

4.3.2 Tests. The first article unit(s) shall be tested as specified in 4.6. Failure of one or more tests shall be cause for rejection of the first article.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination and tests. Sampling shall be in accordance with MIL-STD-105, table I and table II-A.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.6.5. Failure of one or more tests shall be cause for rejection.

4.5 Examination. Each FARE system shall be examined as specified herein for the following defects.

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Major

101. Materials not as specified (see 3.3).
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.3).
105. Use of cadmium plated fasteners not in accordance with stated requirements (see 3.3.4).
106. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the FARE system (see 3.3.5).
107. Any part (or component) not in accordance with the QAP requirements as shown on the drawings (see drawings).
108. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified (see drawings).
109. Threads and threaded assemblies not as specified (see 3.4).
110. Components missing or not as specified (see 3.5).
111. Hardware not as specified (see drawings).
112. Identification marking, incomplete, or illegible (see drawings).
113. Treatment and painting not as specified (see 3.15).
114. Rail transportability not as specified (see 3.16).
115. Instruction plate/placard not as specified (see 3.17).
116. Workmanship not as specified (see 3.18).
117. Welding and welds not as specified (see 3.18.4).
118. Welder/welding operator certification not as specified (see 3.18.5).
119. Human factors engineering not as specified (see 3.20).

4.6 Tests.

4.6.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The FARE shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by the maintenance schedule prepared and submitted by the contractor prior to test. Before performing the following tests, the FARE shall be serviced with military fuels and lubricants designated for use at the ambient temperature at which the test is being conducted. The test fluid shall be gasoline conforming to MIL-G-3056, jet fuel conforming to MIL-T-5624, diesel fuel conforming to MIL-F-46162, solvent conforming to P-D-680, commercial turbine fuel conforming to ASTM D 1635, or kerosine conforming to ASTM D 3699.

4.6.2 Environmental.

4.6.2.1 High temperature. The FARE shall be tested as specified in MIL-STD-810, method 501.3, procedure I and II. The maximum ambient storage temperature shall be 160 °F, and the maximum ambient operating temperature shall be 125 °F. The operating period of procedure I shall be a minimum of 2 hours. The FARE shall be

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operated at rated capacity. Nonconformance to 3.10.1 and 3.10.2 shall constitute failure of this test.

4.6.2.2 Low temperature. The FARE shall be tested as specified in MIL-STD-810, method 502.3, procedure I and II. The minimum ambient storage temperature shall be -50 °F. The minimum ambient operating temperature shall be -25 °F. The operating period of procedure I shall be a minimum 2 hours. The FARE shall be operated at rated capacity. Nonconformance to 3.10.1 and 3.10.2 shall constitute failure of this test.

4.6.3 Reliability. The FARE shall be tested as specified in 4.6.5 for 585 hours. The lower test limit shall be 75 hours and the upper test limit shall be 150 hours. A failure is any malfunction which the operator cannot correct within 30 minutes by adjustment, repair, or replacement action; using controls, on-equipment tools, or parts; and which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Serious damage to the FARE system by continued operation.
c. Serious personnel safety hazards.

Accept if five or fewer failures occur and reject if six or more failures occur. Nonconformance to 3.11 or 3.12 shall constitute failure of this test.

4.6.4 Maintainability.

4.6.4.1 Maintenance support evaluation. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.12. Failure of the support elements to maintain the FARE as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.6.4.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.12 shall constitute failure of this test.

4.6.5 Performance.

4.6.5.1 Hydrostatic. The section of the FARE system that is under pressure during service shall be subjected to a hydrostatic test pressure equal to 150 percent of the working pressure of the system for at least 30 minutes. Any damage or leakage shall be cause for rejection.

4.6.5.2 Capacity. The FARE shall be assembled for operation as specified in the applicable technical manual. The FARE shall be operated for a period of nine continuous hours and flow rate and total head shall be determined at the beginning of each half hour by means of calibrated instruments. Nonconformance to 3.13.1 shall constitute failure of this test.

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4.6.5.3 Compatibility. With the FARE assembled and operating as specified in 4.6.5.1, an inspection of all components and fittings shall be made. Nonconformance to 3.13.2 shall constitute failure of this test.

4.6.5.4 Continuity. With the FARE assembled and operating as specified in 4.6.5.1, an ohmmeter shall be used to determine electrical continuity of the static ground assemblies. Nonconformance to 3.13.3 shall constitute failure of this test.

4.6.6 Rail impact test. The first article FARE, prepared for shipment in accordance with 5.3, shall be tested in accordance with MIL-STD-810 to prove conformance to 3.16.1. Inability to meet the requirements of 3.16.1 shall constitute failure of this demonstration.

4.6.8 Safety. The FARE shall be evaluated throughout testing as specified in 4.6.1. Nonconformance to 3.19 shall constitute failure of this test.

4.6.9 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal horizontal arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 and 3.21 provisions shall constitute failure of this test.

4.7 Inspection comparison. The Government may select FARE at any time during the contract production period and subject these FARE to the examination and tests specified in 4.6. The inspection will be performed by the Government, at a site selected by the Government, on FARE selected at random from those which have been accepted by the Government and will not include the previously inspected first article model and initial production FARE. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison FARE to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison FARE and shall be cause for refusal by the Government to continue acceptance of production FARE until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on FARE previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all FARE accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

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4.8.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, all components that will comprise a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, table I and table II-A.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

120.	Pump assembly not preserved in accordance with the referenced document as specified for level A (see 5.2.1.1).	
121.	Filter-separator and replacement filter elements not preserved in accordance with the referenced document as specified for level A and level C (see 5.2.1.2 and 5.2.2).	
122.	Discharge hoses not drained and open ends not closed or covered as specified for level A (see 5.2.1.3).	
123.	Discharge hoses not coiled and secured in the frames as specified for level A (see 5.2.1.3).	
124.	Nozzles and fittings not placed in the canvas container and secured as specified for level A (see 5.2.1.3).	
125.	Ends of suction hoses and ground rods not cushioned as specified for level A (see 5.2.1.4).	
126.	Suction hoses and ground rods not placed in the canvas container and secured as specified for level A (see 5.2.1.4).	
127.	Consolidation not as specified for level A (see 5.2.1.8).	
128.	Packing not as specified for level A, level B, or level C (see 5.3).	
129.	Marking missing, illegible,incorrect or incomplete for level A, level B, or level C (see 5.4).	

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove, prior to starting production packaging, that applied preservation, packing, and marking comply with the packing requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article model FARE or a production FARE. If the first article model FARE is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model FARE and production FARE.

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5.2 Preservation. Preservation shall be level A or level C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pump assembly. Components of the pump assembly shall be preserved in accordance with level A requirements of MIL-P-52746.

5.2.1.2 Filter-separator and replacement filter elements. The filter-separator and replacement filter elements shall be preserved in accordance with MIL-F-52556.

5.2.1.3 Discharge hose and fitting kits. Discharge hoses shall be drained of fuel or water and shall be coiled flat. Open ends shall be closed with appropriate caps or plugs or shall be covered with material conforming to L-P-378, type I, class 1, thickness 4 mil, secured in place with tape conforming to PPP-T-60, type IV. The coiled hoses shall be secured in the frames with the web straps provided. Nozzles and fittings shall be placed in the canvas container in their designated places and shall be secured by the canvas container flaps and web straps provided.

5.2.1.4 Suction hose and ground rod kits. The suction hoses and ground rods shall be placed in the canvas containers in their designated spaces. The ends of each suction hose and ground rod shall be wrapped with cushioning material to prevent damage to the canvas container (see figure 1). The container flaps and web straps shall be secured. The secured canvas containers shall be placed in a close-fitting box conforming to PPP-B-601, domestic type, style optional, grade B, as shown in figure 1.

5.2.1.5 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.6 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.8 Consolidation. The replacement filter elements, repair parts, maintenance tools and technical publications for each FARE preserved as specified shall be consolidated in a close-fitting box conforming to PPP-B-601, domestic type, style I, grade B. Closure shall be in accordance with the box specification. Strapping is not required. Box configuration shall be such as to permit the packing shown on figure 2 (see 6.11).

5.2.2 Level C. The filter separator and replacement filter elements shall be preserved as specified for level A (see 5.2.1.2). All other components of the FARE shall be preserved in a manner to afford protection against deterioration and damage during shipment from the contractor to the original destination.

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5.3 Packing. Packing shall be level A, level B, or level C as specified (see 6.2).

5.3.1 Level A and level B. The boxed and unboxed components of each complete FARE, preserved as specified in 5.2, shall be packed in a box conforming to PPP-B-601, overseas type, style J. Level B boxes shall be domestic type. The components shall be arranged and secured to the skid-base with U-bolts or equivalent as shown by figure 2, to resist any movement or shifting of the components during testing or shipping. The FARE shall then be protected with the box cover of minimum size required to fit over and completely enclose the components. The box cover shall be secured to the skid base by use of wood members, nails, nuts, and bolts or any combination thereof. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable, except for level B, the strapping may be coated finish. The strapping shall encircle the box girthwise, laterally and longitudinally; those straps perpendicular to the skids shall be placed between the skids and the lower surface of the box bottom panel.

5.3.2 Level C. The components comprising a complete FARE, preserved as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.4 Marking. In addition to any special marking specified in the contract on purchase order, marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The FARE is intended for refueling helicopters in forward operational areas and for refueling military vehicles and powered equipment. The FARE shall be used in all environmental conditions listed herein.

6.2 Acquisition data. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required (see 3.2 and 4.3).
d. When justification is required to use cadmium plating (see 3.3.4).
e. Color required (see 3.15).
f. When the welding process will not be at the option of the contractor (see 3.18.4).
g. Level of preservation and level of packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the items shall be preproduction models. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first

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article test results and disposition of the document's first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 First article pack. Any changes or deviations of production pack from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the FARE in accordance with this specification.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under chapter 2 of MIL-HBDK-113.

6.6 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Definitions.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7.2 Maintenance ratio. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded.

6.8 Cadmium plated fasteners. The contracting officer shall require the contractor to justify the use of cadmium plates fasteners in writing (see 3.3.4). Citing standard DoD or commercial practice will not be considered sufficient

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justification. A specific rationale is required for each specific/different application of cadmium plated fasteners.

6.8.1 Cadmium plated fasteners part numbers. In those instances where cadmium plates fasteners are required and specified (see 3.3.4), drawings and technical manuals shall show these cadmium plated fasteners as separate part numbers to preclude the inadvertent use of zinc coated fasteners.

6.9 Subject term (key word) listing.

Aircraft refueling
Fuel-dispensing system
Ground vehicle refueling
Powered equipment refueling
Refueling system

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

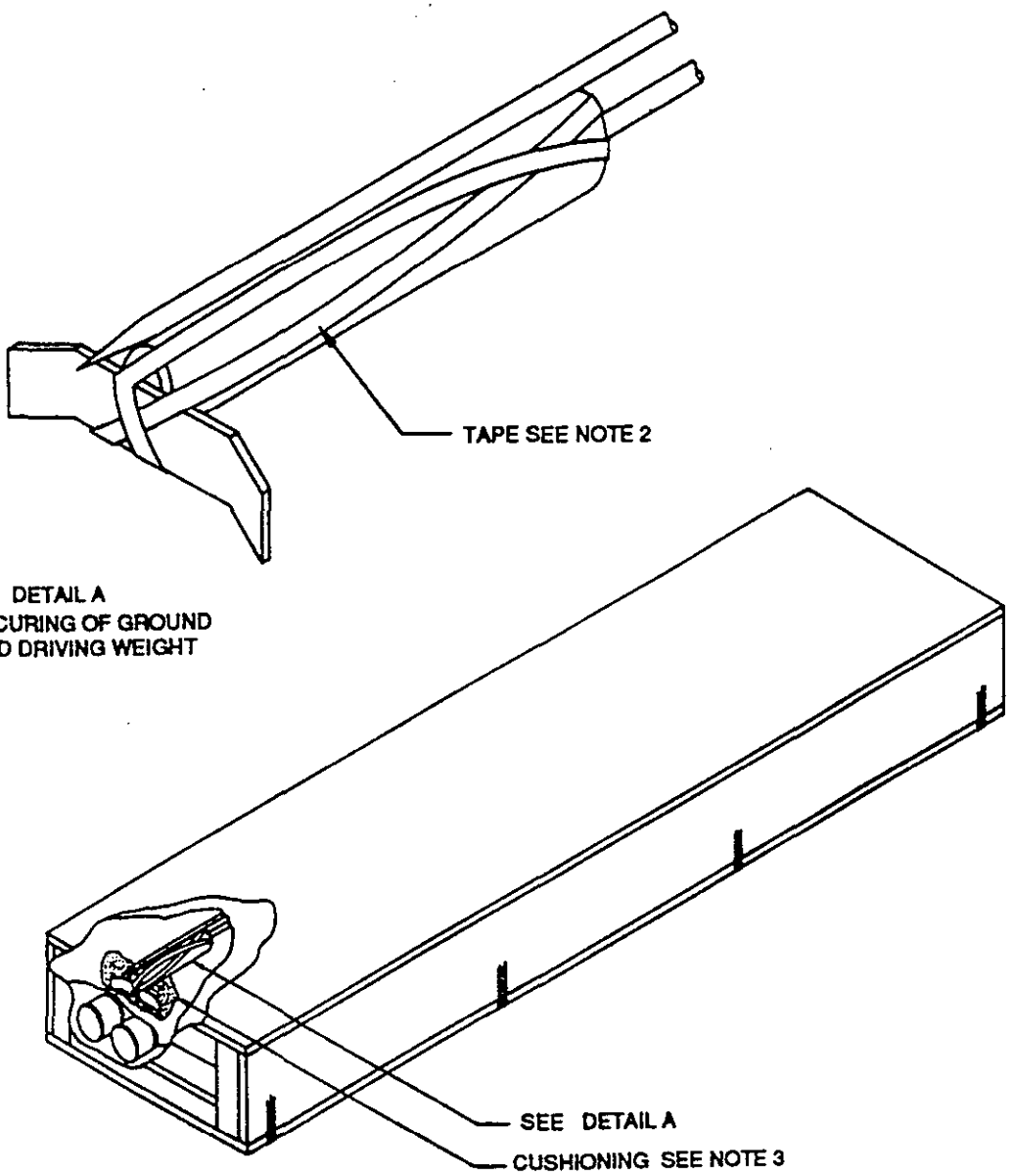
6.11 Figure 2. Figure 2 is for information only.

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Preparing activity:
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Project 4930-A368

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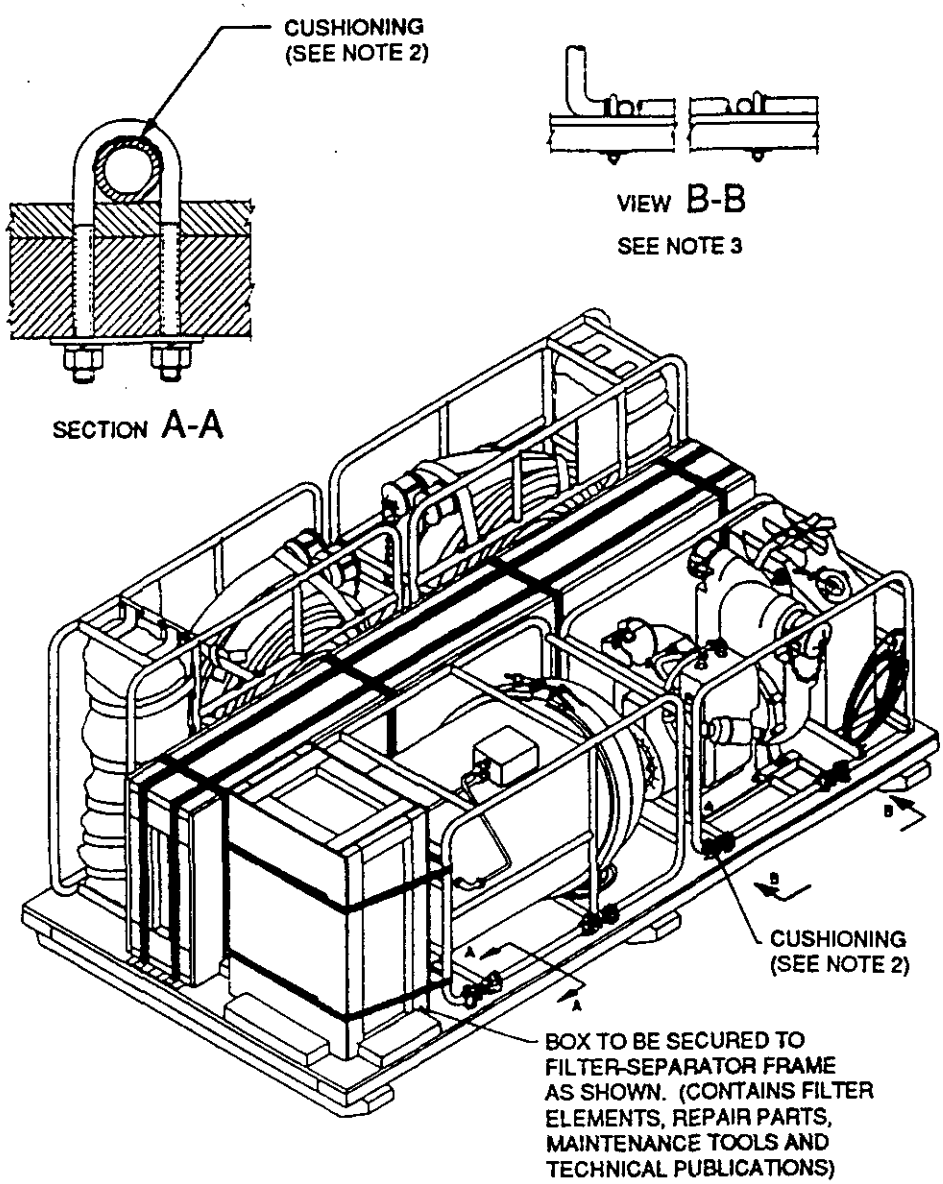


NOTES:

1. CUSHIONING, SHALL BE PROVIDED BETWEEN BOX ENDS AND CONTENTS. (SUCTION HOSES AND GROUND RODS).
2. GROUND RODS WILL BE SECURED WITH TAPE CONFORMING TO PPP-T-97, TYPE VI.
3. ENDS OF GROUND RODS SHALL BE WRAPPED WITH CUSHIONING CONFORMING TO PPP-C-843 BEFORE PLACING THE GROUND RODS IN THE CANVAS CONTAINER. CUSHIONING SHALL BE SECURED WITH TAPE, PPP-T-97, TYPE VI.

FIGURE 1. Box details.

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NOTES:

1. IN CENTER AREAS WHERE U-BOLTS WILL NOT SECURE THROUGH A CLEAT MEMBER OF THE BASE, SLEEPERS OF NOMINAL 2 INCH THICK LUMBER SHALL BE PROVIDED.
2. CUSHIONING, A-A-1899, TYPE II, CLASS A, SHALL BE PROVIDED UNDER ALL U-BOLTS AS SHOWN.
3. U-BOLTS SHALL BE POSITIONED ON ALL FRAME MEMBERS AS SHOWN TO RESIST MOVEMENT IN EITHER DIRECTION.

FIGURE 2. Packing Arrangement.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-52749C(ME)	2. DOCUMENT DATE (YYMMDD) 920212

3. DOCUMENT TITLE
Forward Area Refueling Equipment (FARE)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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